SAGAR PUBLIC SCHOOL

Integrated Programme • **Topic: Work, Power & Energy**

JEE MAIN Practice Sheet – 2

Name: _____ **Class / Sec:** _____ **Date:** _____

Time Allowed: 60 Mins • **Total Questions: 20** • 12 from JEE Main PYQs + 8 from Module

A Energy & Conservation (Q1–8) **B** Work–Energy Theorem (Q9–14) **C** Power (Q15–17) **D** Vertical Circular Motion (Q18–20)

SECTION A – Energy & Application of Energy Conservation (Q1–Q8)

Q1. *[PYQ · AIEEE 2005]*

A spherical ball of mass 20 kg is stationary at the top of a hill of height 100 m. It rolls down a smooth surface to the ground, then climbs up another hill of height 30 m and finally rolls down to a horizontal base at a height of 20 m above the ground. The velocity attained by the ball is

(A) 20 m/s (B) 40 m/s (C) $10\sqrt{30}$ m/s (D) 10 m/s

Q2. *[Module · Ex. S-1, Q27]*

The potential function for a conservative force is given by $U = k(x+y)$. Find the work done by the conservative force in moving a particle from the point A$(1, 1)$ to point B$(2, 3)$.

Q3. *[PYQ · JEE Main 2024, 4 Apr Morning]*

If a rubber ball falls from a height h and rebounds upto the height of $h/2$. The percentage loss of total energy of the initial system as well as velocity of ball before it strikes the ground, respectively, are:

(A) 50%, $\sqrt{2gh}$ (B) 50%, $\sqrt{gh}$

(C) 50%, $\sqrt{\dfrac{gh}{2}}$ (D) 40%, $\sqrt{2gh}$

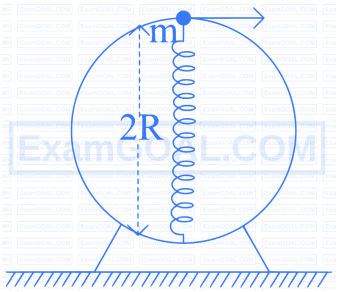
Q4. *[Module · Ex. O-2, Q1]*

A light spring of length 20 cm and force constant 2 N/cm is placed vertically on a table. A small block of mass 1 kg falls on it. The length h from the surface of the table at which the block will have the maximum velocity is:

(A) 20 cm (B) 15 cm (C) 10 cm (D) 5 cm

Q5. *[PYQ · JEE Main 2025, 28 Jan Morning]*

A bead of mass m slides without friction on the wall of a vertical circular hoop of radius R as shown in figure. The bead moves under the combined action of gravity and a massless spring (k) attached to the bottom of the hoop. The equilibrium length of the spring is R. If the bead is released from top of the hoop with (negligible) zero initial speed, velocity of bead, when the length of spring becomes R, would be (spring constant is k, g is acceleration due to gravity)

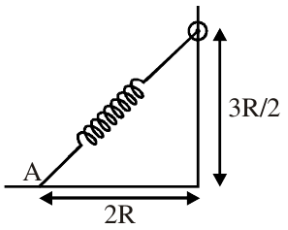


(A) $\sqrt{2Rg + \dfrac{kR^2}{m}}$ (B) $\sqrt{3Rg + \dfrac{kR^2}{m}}$

(C) $\sqrt{2Rg + \dfrac{4kR^2}{m}}$ (D) $2\sqrt{gR + \dfrac{kR^2}{m}}$

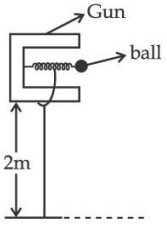
Q6. *[Module · Ex. S-2, Q2]*

A ring of mass m can slide over a smooth vertical rod. The ring is connected to a spring of force constant $K = \dfrac{4mg}{R}$ where $2R$ is the natural length of the spring. The other end of the spring is fixed to the ground at a horizontal distance $2R$ from the base of the rod. The mass is released at a height of $1.5R$ from ground.
(i) Calculate the work done by the spring. (ii) Calculate the velocity of the ring as it reaches the ground.



Q7. *[PYQ · JEE Main 2021, 20 Jul Morning]*

In a spring gun having spring constant 100 N/m a small ball B of mass 100 g is put in its barrel (as shown in figure) by compressing the spring through 0.05 m. There should be a box placed at a distance d on the ground so that the ball falls in it. If the ball leaves the gun horizontally at a height of 2 m above the ground. The value of d is _____ m. $(g = 10\,\text{m/s}^2)$



Q8. *[PYQ · JEE Main 2021, 1 Sep Evening]*

An engine is attached to a wagon through a shock absorber of length 1.5 m. The system with a total mass of 40,000 kg is moving with a speed of 72 km h^{-1} when the brakes are applied to bring it to rest. In the process of the system being brought to rest, the spring of the shock absorber gets compressed by 1.0 m. If 90% of energy of the wagon is lost due to friction, the spring constant is _____$\times 10^5$ N/m.

SECTION B – Work–Energy Theorem (Q9–Q14)

Q9. *[PYQ · JEE Main 2026, 21 Jan Evening]*

A body of mass 2 kg is moving along x-direction such that its displacement as function of time is given by $x(t) = \alpha t^2 + \beta t + \gamma$ m, where $\alpha = 1$ m/s^2, $\beta = 1$ m/s and $\gamma = 1$ m. The work done on the body during the time interval $t = 2$ s to $t = 3$ s, is _____ J.

(A) 42 (B) 24 (C) 49 (D) 12

Q10. *[Module · Ex. S-1, Q10]*

The net force exerted on a particle acts in the $+x$ direction. Its magnitude increases linearly from zero at $x = 0$, to 24.0 N at $x = 3.0$ m. It remains constant at 24.0 N from $x = 3.0$ m to $x = 8.0$ m and then decreases linearly to zero at $x = 13.0$ m. Determine the work done to move the particle from $x = 0$ to $x = 13.0$ m.

Q11. *[PYQ · JEE Main 2022, 25 Jul Morning]*

A body of mass 0.5 kg travels on straight line path with velocity $v = (3x^2 + 4)$ m/s. The net workdone by the force during its displacement from $x = 0$ to $x = 2$ m is:

(A) 64 J (B) 60 J (C) 120 J (D) 128 J

Q12. *[Module · Ex. O-1, Q6]*

A bicyclist comes to a skidding stop in 10 m. During this process, the force on the bicycle due to the road is 200 N and is directly opposed to the motion. The work done by the cycle on the road is

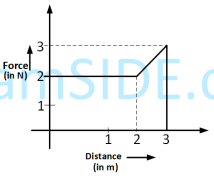
(A) +2000 J (B) −200 J (C) zero (D) −20,000 J

Q13. *[PYQ · JEE Main 2023, 31 Jan Morning]*

A lift of mass $M = 500$ kg is descending with speed of 2 m s^{-1}. Its supporting cable begins to slip thus allowing it to fall with a constant acceleration of 2 m s^{-2}. The kinetic energy of the lift at the end of fall through to a distance of 6 m will be _____ kJ.

Q14. *[PYQ · JEE Main 2019, 8 Apr Morning]*

A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3 m is:



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(A) 6.5 J (B) 2.5 J (C) 5 J (D) 4 J

SECTION C – Power (Q15–Q17)

Q15. *[PYQ · JEE Main 2022, 27 Jul Morning]*

Sand is being dropped from a stationary dropper at a rate of 0.5 kg s^{-1} on a conveyor belt moving with a velocity of 5 m s^{-1}. The power needed to keep the belt moving with the same velocity will be:

(A) 1.25 W (B) 2.5 W (C) 6.25 W (D) 12.5 W

Q16. *[Module · Ex. S-2, Q1]*

Water is pumped from a depth of 10 m and delivered through a pipe of cross section 10^{-2} m^2 upto a height of 10 m. If it is needed to deliver a volume 0.2 m^3 per second, find the power required. [Use $g = 10$ m/s^2]

Q17. *[PYQ · JEE Main 2020, 5 Sep Evening]*

A body of mass 2 kg is driven by an engine delivering a constant power of 1 J/s. The body starts from rest and moves in a straight line. After 9 seconds, the body has moved a distance (in m) _____.

SECTION D – Vertical Circular Motion (Q18–Q20)

Q18. *[PYQ · JEE Main 2024, 27 Jan Evening]*

A ball suspended by a thread swings in a vertical plane so that its magnitude of acceleration in the extreme position and lowest position are equal. The angle (θ) of thread deflection in the extreme position will be:

(A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) $2\tan^{-1}\left(\frac{1}{2}\right)$

(C) $2\tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$ (D) $\tan^{-1}\left(\sqrt{2}\right)$

Q19. *[Module · Ex. O-1, Q30]*

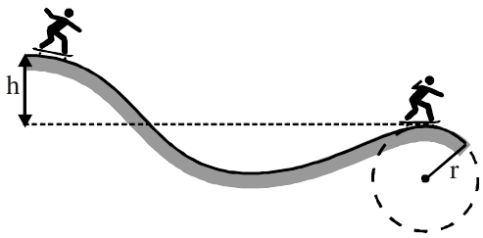
A stone tied to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position, and has a speed u. The magnitude of the change in its velocity as it reaches a position where the string is horizontal is –

(A) $\sqrt{u^2 - 2gL}$ (B) $\sqrt{2gL}$

(C) $\sqrt{u^2 - 2gL}$ (D) $\sqrt{2\left(u^2 - gL\right)}$

Q20. *[Module · Ex. S-1, Q30]*

A skier starts from rest at the top of a hill. The skier coasts down the hill and up a second hill, as the drawing illustrates. The crest of the second hill is circular, with a radius of $r = 36$ m. Neglect friction and air resistance. What must be the height h (in m) of the first hill so that the skier just loses contact with the snow at the crest of the second hill?



SELF ASSESSMENT (to be filled by the student after the test)

Time Taken: _____ min **Attempted:** _____ / 20 **Unattempted:** _____

Correct: _____ **Incorrect:** _____ **Accuracy:** _____ %

Section	Questions	Correct	Incorrect	Topics / concepts I must revise
A – Energy & Conservation	8			
B – Work–Energy Theorem	6			
C – Power	3			
D – Vertical Circular Motion	3			

Personal Feedback (what went well / what went wrong / silly mistakes / plan for next sheet):

_____

_____

_____

_____

Student's Signature: _____ **Teacher's Remark / Signature:** _____